Inorganic Cume
Feb. 12, 1994

Acid - Base Models and Chemistry

1) Write balanced chemical equations showing PF_3 acting as a Lewis acid and as a Lewis base. Predict the most probable hybridization for P in the product for both cases.

2) Draw structural formulas for the following :

H_2SO_5 , $H_4P_2O_7$, $H_2S_4O_6$, periodic acid

3) Pauling has suggested relationships for predicting the approximate value of the pKa for oxyacids [pKa ≈ 8 -5(m-n) and pKa(i+1) - pKa(i) ≈ 5]. Define the values of m and n and in one sentence each indicate why the general form of the equations seems reasonable. Use these relationships to predict the values of the successive pKa's for H_2SO_4 and H_2SO_3.

4) Phosphorous acid is a *diprotic* acid with pKa value 1.8 and 6.6. Can this observation be consistent with the Pauling equation?

5) The K_{eq} values for the reactions below are *both* much greater than one. How can you rationalize Al(III) being the stronger acid in the first reaction and the weaker acid in the second?

$$Al^{3+} + Hg_2F^{1+} \leftrightarrows AlF^{2+} + Hg_2^{\ 2+}$$

$$AlI^{2+} + Hg_2^{\ 2+} \leftrightarrows Al^{3+} + Hg_2F^{1+}$$

6) The gas phase proton affinities of F^- and Me_3N are 1554 and 884 kJ-mol^{-1} respectively. Comment on the probable magnitude of the equilibria (e.g. K >1 ,<1 ,...) and any differences you expect.

$$HF\ (g) + Me_3N\ (g) \leftrightarrows Me_3NH^+\ (g) + F^-(g)$$

$$HF\ (aq) + Me_3N\ (aq) \leftrightarrows Me_3NH^+\ (aq) + F^-(aq)$$